Per-DOF Grey-Box Identification of an Omnidirectional Soccer Robot from Closed-Loop Competition Logs: A Multi-Horizon Fidelity Study against an ODE Simulator

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***Abstract*—**

***Index Terms*—digital twin, system identification, grey-box modeling, omnidirectional robot, robot soccer, sim-to-real, fidelity evaluation**

I. Introduction

Contributions.

- **C1.** A per-DOF grey-box (FOPDT) identification methodology for an omnidirectional robot from *closed-loop* competition logs, with per-DOF hold-out validation.
- **C2.** A free-run, multi-horizon, re-anchored fidelity protocol with motion-stratified windows and a statistically gated verdict (paired non-overlapping Wilcoxon + moving-block bootstrap + effect size).
- **C3.** An explicit, validated comparison showing the per-DOF twin attains higher *translational* fidelity than rSim on translation-dominated windows.
- **C4.** A characterization of the *non-identifiability of yaw* (ω) dynamics from closed-loop data, treated as a finding (not a failure) that bounds heading fidelity and motivates dedicated rotation data.

II. Related Work

III. Method

A. Per-DOF FOPDT twin

We model the robot as three decoupled single-input single-output channels, one per degree of freedom (DOF) $i \in \{v_x, v_y, \omega\}$, each a first-order-plus-dead-time (FOPDT) transfer function from commanded to attained body velocity,

$$G_i(s) \;=\; \frac{K_i\, e^{-T_{di}s}}{\tau_i s + 1}, \tag{1}$$

with per-DOF gain K_i, time constant τ_i, and transport delay T_{di}. The body-frame commands v_x, v_y (longitudinal/lateral, $\mathrm{m/s}$) and ω (yaw rate, $\mathrm{rad/s}$) are the inputs; the outputs are the attained body velocities. The full parameter vector has fixed order $[K_{vx}, \tau_{vx}, T_{dvx}, K_{vy}, \tau_{vy}, T_{dvy}, K_{\omega}, \tau_{\omega}, T_{d\omega}]$ (nine parameters). This is a *behavioral* (observational) model: it targets the input-to-pose response, not a white-box description of wheel/contact mechanics.

B. Discretization and pose integration

The twin is simulated in discrete time on a uniform grid with step $\Delta t = 1/60\,\mathrm{s}$. For each DOF the command is first delayed, then passed through the first-order lag, then scaled by the gain. The transport delay is a zero-order-hold (ZOH) integer shift of $m_i = \lceil T_{di}/\Delta t \rceil$ samples,

$$u_i^d[k] \;=\; u_i[k - m_i]\,, \qquad m_i = \lceil T_{di}/\Delta t \rceil, \tag{2}$$

with the convention $u_i \equiv 0$ before the segment start. The first-order lag uses the exact ZOH update

$$v_i[k] \;=\; v_i[k-1] + \big(1 - e^{-\Delta t/\tau_i}\big)\,\big(u_i^d[k] - v_i[k-1]\big), \tag{3}$$

and the attained body velocity is $\hat{v}_i[k] = K_i\, v_i[k]$ (for $\tau_i \to 0$ the lag is instantaneous). The body velocity is integrated to the field (world) frame through the current heading using the trapezoidal rule,

$$\theta[k] = \theta[k-1] + \tfrac{\Delta t}{2}\big(\hat{v}_\omega[k] + \hat{v}_\omega[k-1]\big), \tag{4}$$

$$x[k] = x[k-1] + \tfrac{\Delta t}{2}\big(g_x[k] + g_x[k-1]\big), \tag{5}$$

$$y[k] = y[k-1] + \tfrac{\Delta t}{2}\big(g_y[k] + g_y[k-1]\big), \tag{6}$$

with $g_x = \cos\theta\,\hat{v}_x - \sin\theta\,\hat{v}_y$ and $g_y = \sin\theta\,\hat{v}_x + \cos\theta\,\hat{v}_y$. This discrete plant is a faithful port of the reference analytic dead-reckoning implementation; an independent reconciliation against the high-fidelity Simulink model agrees to $\leq 6\,\mathrm{mm}$ in position and $\leq 9 \times 10^{-4}\,\mathrm{rad}$ in heading on a reference event.

C. Data and pre-processing

The data are closed-loop competition logs of a single omnidirectional 4-wheeled SSL robot, decoded from the league protocol stream into per-event commands and pose at $\sim$60 Hz. Commands (v_x, v_y in $\mathrm{m/s}$, ω in $\mathrm{rad/s}$) are in the *body* frame; pose (x, y in m after $\mathrm{mm}\to\mathrm{m}$ conversion, heading in rad, unwrapped) is in the *field* frame. Measured body velocity, used only for identification and for the initial condition, is derived from the vision position by a Savitzky–Golay derivative rotated into the body frame. The native time grid is irregular (command bursts); for identification the signals are resampled to $\Delta t = 1/60\,\mathrm{s}$, while the fidelity metric uses time-based windows on the native grid. The dataset comprises 46 events across four motion labels (`front_to_back`,

`side_to_side`, `shoot_to_goal`, `square`). A frozen list of 10 events is held out for evaluation and is never used for fitting.

D. Identification

Each translational DOF is identified independently as a P1D (FOPDT) model by prediction-error minimization, with an event-level hold-out (seed 42). The excitation diagnostic selects the cleanest source label per DOF: v_x from `front_to_back` and v_y from `side_to_side` (command-to-velocity cross-correlation ≈ 0.99). The yaw channel ω is treated separately. A direct P1D fit on the highest-rotation closed-loop label fails validation (Sec. IV-D): this is not a property of the DOF but of the excitation available in competition logs, so we identify ω from a dedicated open-loop collection instead (Sec. III-E), and both the closed-loop attempt and the open-loop result are reported for contrast.

E. Dedicated open-loop yaw excitation

Because closed-loop competition commands under-excite ω (Sec. V), we collected a dedicated, single-session, single-robot open-loop dataset with $v_x = v_y = 0$ throughout: an alternating-sign staircase of commanded yaw-rate amplitudes $\{0.25, 0.5, 1, 2.5, 5\}\,\mathrm{rad/s}$, held for $\sim 10\,\mathrm{s}$ per step (ten events total). The two lowest amplitudes ($\pm 0.25, \pm 0.5$) never engage the motors (zero commanded PWM) and are used only to characterize the electrical per-wheel dead-zone (Sec. IV-D); the remaining six events ($\pm 1, \pm 2.5, \pm 5\,\mathrm{rad/s}$) form the linear regime used for identification, split event-wise into four training and two hold-out events (the same P1D/prediction-error pipeline, resampling, and Savitzky–Golay pre-processing as v_x, v_y). Given the small sample (six events in the linear regime), the acceptance criteria — physical parameter bounds, hold-out fit $\geq 60\,\%$, and the multi-horizon $\mathrm{SE}(2)$ gate against the closed-loop placeholder — were fixed and committed *before* estimation, as a methodological safeguard against post-hoc adjustment of the criteria to the result.

F. Free-run replay and multi-horizon metric

Fidelity is measured in *free-run*: starting from the true pose at a window anchor t_0, the twin is driven only by the command stream and never re-anchored to the real pose inside the window (no teacher forcing). Teacher forcing is explicitly forbidden because it would measure only single-step noise and inflate the apparent fidelity.

A segment is cut into overlapping windows at a fixed stride; each window is re-anchored to the true $\mathrm{SE}(2)$ pose at its anchor t_0 and rolled out to a maximum horizon $H_{\max} = 90$ samples (1.5 s). At horizons h we report two errors *separately*:

$$e_{\mathrm{pos}}(w,h) = \left\| (x_{\mathrm{sim}} - x_{\mathrm{real}},\, y_{\mathrm{sim}} - y_{\mathrm{real}}) \right\|_2, \tag{7}$$

$$e_{\mathrm{ang}}(w,h) = \left| \,\mathrm{wrap}_\pi(\theta_{\mathrm{sim}} - \theta_{\mathrm{real}}) \,\right|, \tag{8}$$

evaluated at $h \in \{6, 12, 24, 42, 60, 90\}$ samples ($\{0.1, 0.2, 0.4, 0.7, 1.0, 1.5\}$ s), with reference horizon $h_{\mathrm{ref}} = 24$ (0.4 s). Keeping e_{pos} and e_{ang} separate is deliberate: e_{pos} is the claim of this paper, whereas e_{ang} exposes the limit imposed by the population of rotation-rich windows available for evaluation (Sec. IV-B), independently of whether ω itself is identified.

a) Motion stratification: Each window is binned by its rotational fraction $r = a_{\mathrm{rot}}/(a_{\mathrm{rot}} + a_{\mathrm{trans}})$, where a_{trans} and a_{rot} accumulate translational command magnitude and yaw rate scaled by the robot radius. Windows are *translation-dominated* ($r < 0.2$), *mixed* ($0.2 \leq r \leq 0.5$), or *rotation-dominated* ($r > 0.5$).

b) Statistical gate: Curves use all (dense) windows; inference uses only *non-overlapping* windows (stride $\geq H_{\max}$) for a paired one-sided Wilcoxon signed-rank test ($e_{\mathrm{pos}}^{\mathrm{twin}} < e_{\mathrm{pos}}^{\mathrm{baseline}}$), plus a moving-block bootstrap (block $\sim H_{\max}$) over the dense windows for the gap confidence interval; both are reported and any sign/order divergence is flagged. Effect size is the rank-biserial correlation of the paired differences. A bin receives a GO verdict only if, at all $h \geq h_{\mathrm{ref}}$, the gap is positive and non-decreasing, the Wilcoxon $p < \alpha$ ($\alpha = 0.05$), the effect size ≥ 0.2, and the bootstrap CI excludes zero, subject to a minimum of 10 independent windows. Mixed and rotation-dominated bins are returned as INDETERMINATE whenever the independent-window count falls below $N_{\min}$, which is the case throughout this paper (Sec. IV-B).

c) Horizon grid and anchor convention: All results in this paper use the $\mathrm{SE}(2)$ (full-pose) re-anchor described above. Two independently implemented evaluation harnesses are used, however, each with its own horizon grid: the translational fidelity results (Sec. IV-C, twin vs. ideal/rSim) use $h \in \{0.1, 0.2, 0.4, 0.7, 1.0, 1.5\}$ s as given above, while the yaw gate (Sec. IV-D) reuses an earlier per-DOF campaign harness with $h \in \{0.5, 1, 2, 2.5, 4\}$ s. We report each result on its native grid rather than recompute either, and flag which grid is in use at each point.

G. Baselines

We compare the twin against two baselines under the *same* windows, horizons, and gate. The first is an *ideal-kinematics* plant in which the attained body velocity equals the command instantaneously (no gain, lag, or delay), integrated by the same trapezoidal scheme. The second is **rSim**, the ODE-based physics engine of the rSoccer environment, driven by the same body-frame commands with motion control disabled so that the engine dynamics are isolated. rSim provides no velocity setter at reset (the robot starts from rest), so two initial-condition fairness modes are used: **Mode A** (primary) warms the engine up over $L_w = 15$ samples with the real pre-anchor commands and then applies a rigid $\mathrm{SE}(2)$ re-anchor at t_0, mirroring how the twin is seeded with the real body velocity; **Mode B** (robustness) uses no warm-up and discards the first 15 samples for *both* plants. The twin's transport delay has no counterpart in rSim, which is precisely where the twin is expected to win at short horizons.

TABLE I

IDENTIFIED PER-DOF FOPDT PARAMETERS AND HOLD-OUT FIT. ALL THREE CHANNELS ARE IDENTIFIED (NO PLACEHOLDER); ω USES A DEDICATED OPEN-LOOP ROTATION-ONLY COLLECTION (SEE SEC. IV-D), NOT THE CLOSED-LOOP COMPETITION LOGS USED FOR v_x, v_y.

DOF	K	τ (s)	T_d (s)	hold-out fit
v_x	1.119	0.072	0.069	86.7 %
v_y	1.052	0.127	0.056	87.1 %
ω	1.002	0.083	0.089	71.1 %

IV. RESULTS

A. Identified model

Table I reports the per-DOF parameters. Both translational channels re-identified as fast FOPDT models and validated on held-out events, with hold-out fit 86.7% (v_x) and 87.1% (v_y). Under the multi-horizon SE(2) gate against the previous (slower) translational parameters, the re-identified models improve every event-by-horizon cell (v_x: $20/20$; v_y: $65/65$) including on the hold-out, roughly halving the median position error at $2.5\,\mathrm{s}$ (e.g., v_y: 0.141 vs. $0.334\,\mathrm{m}$). The yaw channel ω is also identified, from a separate open-loop dataset, with hold-out fit 71.1%; the reasons for the separate data source, and the contrast with the failed closed-loop attempt, are in Sec. IV-D. All three channels are deterministic bit-for-bit under re-execution (identical seed, data, and toolbox version): re-running the identification pipeline reproduces all nine parameters to 0.0 absolute difference.

B. Motion distribution of the hold-out

The hold-out windows are overwhelmingly translation-dominated: of 265 dense windows (49 independent), 264 fall in the translation bin (99.6%), one is rotation-dominated, and none are mixed. This conditions the analysis: translational fidelity is the statistically powered claim, while the rotation and mixed bins lack the population to support a verdict (Fig. 1). This population is unaffected by the yaw model: re-running the bin classification and the twin-vs-ideal gate after ω was identified (Table I) reproduces the same distribution ($264/1/0$) and leaves mixed and rotation-dominated NO POWER (N_{indep}=0), now for lack of rotation-rich competition windows rather than for lack of a yaw model. The rotational fidelity of the twin therefore remains unverified against a competition-like distribution, independently of whether ω is identified (see Limitations).

C. Twin vs. rSim: translational fidelity

On translation-dominated windows (Mode A), the twin's median position error is *below* rSim's at every horizon, and the gap *grows* with the horizon (Table II, Fig. 2). At the reference horizon $h_{\mathrm{ref}} = 0.4\,\mathrm{s}$ the gap is $+0.093\,\mathrm{m}$ (e_{pos}: 0.038 vs. $0.131\,\mathrm{m}$), with paired non-overlapping Wilcoxon $p = 4.4 \times 10^{-11}$ and rank-biserial effect $+0.922$ ($N_{\mathrm{indep}} = 49$). Across horizons the effect size ranges from $+0.731$ ($1.5\,\mathrm{s}$) to $+0.980$ ($0.2\,\mathrm{s}$); the moving-block bootstrap CI excludes zero at every horizon and agrees in sign and order with the Wilcoxon test (no divergence flag), so the bin receives a GO verdict.

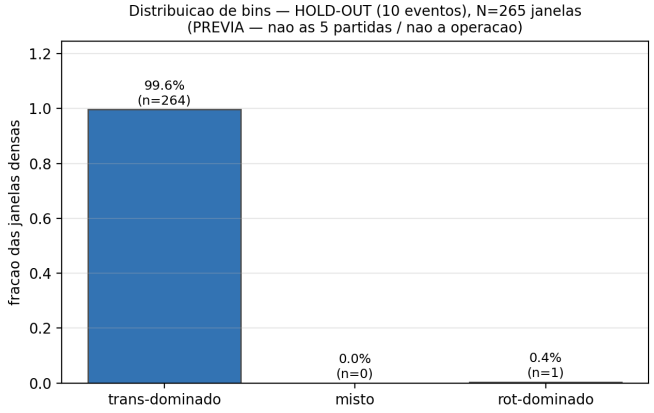


Fig. 1. Motion-content distribution of the hold-out windows; the population is almost entirely translation-dominated.

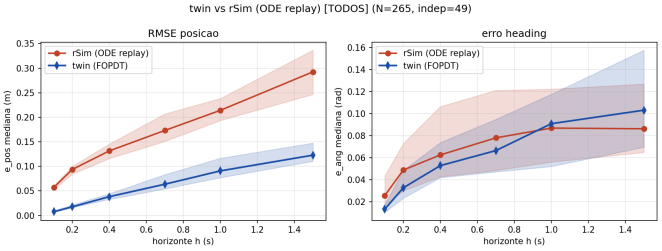


Fig. 2. Free-run fidelity vs. horizon, twin (FOPDT) vs. rSim (ODE engine of rSoccer), Mode A. Left: position error e_{pos} (the claim). Right: heading error e_{ang} on this (translation-dominated) population, generated with the yaw placeholder in place before ω was identified (Sec. IV-D); not re-run with the identified ω. Bands are moving-block bootstrap 95% CIs.

a) Robustness (Mode B): Without warm-up, both plants start from rest and the first 15 samples are dropped for both; rSim's penalty is larger, so the gap widens (e.g., $+0.282\,\mathrm{m}$ at h_{ref}, $p = 4.4 \times 10^{-14}$, effect $+0.987$), and the translation bin remains GO. Mode A is therefore the conservative setting and is used as primary.

b) Twin vs. ideal kinematics: Against the ideal-kinematics baseline the twin is also GO on translation, with a smaller but significant margin (gap $+0.063\,\mathrm{m}$ at h_{ref}, $p = 8.5 \times 10^{-7}$, effect $+0.736$): identified gain/lag/delay help most at short horizons, where ideal kinematics has no transport delay.

c) Heading error: Position and heading errors are reported separately (Fig. 2, right panel). On the translation-dominated population the heading error is small but does *not* carry a verdict, because the rotation and mixed bins lack the independent-window population to be scored (Sec. IV-B) regardless of the yaw model now in use.

D. Yaw identifiability is a function of excitation regime

A direct P1D fit of the yaw channel on high-rotation *closed-loop* competition events gives $(K_\omega, \tau_\omega, T_{d\omega}) \approx$

TABLE II

TWIN (FOPDT) VS. RSIM (ODE) ON TRANSLATION-DOMINATED HOLD-OUT WINDOWS, MODE A (WARM-UP + RIGID SE(2) RE-ANCHOR). e_{pos} MEDIANS IN METERS; gap $= e_{\text{pos}}^{\text{rSim}} - e_{\text{pos}}^{\text{twin}}$; PAIRED NON-OVERLAPPING WILCOXON p; RANK-BISERIAL EFFECT. $N_{\text{indep}} = 49$.

h (s)	twin e_{pos}	rSim e_{pos}	gap (m)	p	effect
0.10	0.0077	0.0572	+0.0495	1.6×10^{-13}	+0.979
0.20	0.0177	0.0935	+0.0758	1.2×10^{-13}	+0.980
0.40	0.0382	0.1312	+0.0930	4.4×10^{-11}	+0.922
0.70	0.0644	0.1731	+0.1087	8.2×10^{-9}	+0.838
1.00	0.0919	0.2144	+0.1225	1.3×10^{-9}	+0.871
1.50	0.1229	0.2927	+0.1699	1.0×10^{-6}	+0.731

$(1.35, 0.035, 0.106)$ with hold-out fit 52.6% (below the 60% acceptance threshold), and using it *worsens* free-run position because the gain > 1 amplifies accumulated heading drift; it is not promoted. A per-amplitude gain analysis (Fig. 3) shows a flat effective gain of ≈ 1.3–1.5 across the closed-loop command range, with no dead-zone knee visible in the *body*-frame yaw signal. A linear re-identification restricted to the high-rotation closed-loop regime fails whiteness (Ljung–Box rejects) and is NO-GO on the multi-horizon gate: it is *worse* than the placeholder at every horizon, pooled gap $+0.0489\,\text{m}$ at $4\,\text{s}$, $p = 3.8 \times 10^{-9}$.

a) The same structure, different data: Firmware inspection locates the dominant non-linearity as an *electrical, per-wheel* dead-zone applied after the inverse kinematics (Sec. III-E), not a body-frame Hammerstein dead-zone that a closed-loop fit could absorb. Re-deriving the commanded-yaw dead-zone from this per-wheel threshold and the robot's wheel-to-contact-radius ratio (measured from CAD, cross-checked against the firmware's implied constant) places it at 0.58–$0.60\,\text{rad/s}$ of body yaw rate – reported as an interval, never a point estimate, because the two independent geometric sources agree only to that precision. Recomputing the fraction of *commanded* yaw-rate samples in the competition logs that fall below this corrected threshold gives 94.3% pooled across all 46 splits, and 77.3% even on `shoot_to_goal`, the label richest in rotation. The closed-loop dataset simply does not excite the rotational mode enough to identify it.

We therefore identify ω from the dedicated open-loop collection (Sec. III-E), using the *same* P1D structure and identification tooling. The result is $K_\omega = 1.0016$ (95 % CI $[0.9993, 1.0040]$), $\tau_\omega = 0.0828\,\text{s}$, $T_{d\omega} = 0.0890\,\text{s}$, with hold-out fit 71.1% (65.0% / 77.2% per event), comfortably above the pre-registered 60% gate; the correlation between τ and T_d is moderate ($\rho = -0.47$), below the 0.8 alert threshold. On the multi-horizon SE(2) gate against the same placeholder, evaluated on the same closed-loop `shoot_to_goal` population used to fail the closed-loop candidate, the open-loop model is *better* at all five horizons, pooled gap $-0.1138\,\text{m}$, $p = 1.7\times10^{-16}$ – the sign of the verdict inverts relative to the closed-loop attempt on the identical gate. Table III contrasts the two rows directly.

b) Interpretation: correlation bias, not plant gain: $K_\omega = 1.0016$ is statistically indistinguishable from unity, and refutes

TABLE III

YAW (ω) P1D IDENTIFICATION: CLOSED-LOOP COMPETITION LOGS VS. A DEDICATED OPEN-LOOP ROTATION-ONLY COLLECTION. SAME MODEL STRUCTURE (EQ. 1) AND IDENTIFICATION TOOLING; ONLY THE EXCITATION REGIME DIFFERS. SE(2) GATE: POOLED MEDIAN GAP VS. PLACEHOLDER (K=1, τ=0.1 s, T_d=0), NON-OVERLAPPING WILCOXON p.

	K_ω	τ_ω (s)	$T_{d\omega}$ (s)	hold-out fit
Closed-loop	1.307	0.0222	0.1179	*n/a*†
Open-loop (dedicated)	**1.0016**	**0.0828**	**0.0890**	**71.1 %**
	gate Δmedian (pool)		gate p	
Closed-loop	+0.0489 m (worse)		3.8×10^{-9}	
Open-loop (dedicated)	$\mathbf{-0.1138}$ m (better)		$\mathbf{1.7 \times 10^{-16}}$	

†The closed-loop candidate was scored by the earlier regime-split criterion (fit 31.9 % high-amplitude / −11.2 % low-amplitude), not a single hold-out number; it failed the physical/whiteness checks regardless (see Sec. IV-D).

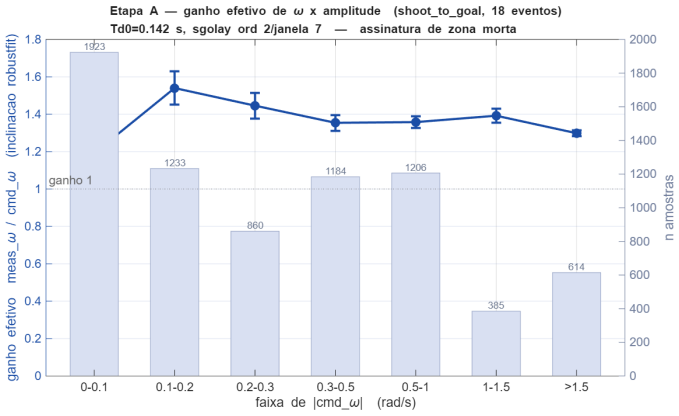


Fig. 3. Effective yaw gain vs. command amplitude in the closed-loop competition logs: flat ≈ 1.3–1.5 with no dead-zone knee visible in the body signal – consistent with a per-wheel, not body-frame, dead-zone, and with 94.3% of commanded amplitudes falling below it.

the closed-loop estimate $K_\omega \approx 1.3$–1.35 as a description of the plant. Under closed-loop control the commanded yaw rate correlates with unmodeled disturbances (surface friction, contact, tactical correction) in a way that inflates the apparent gain; this is a property of the *data regime*, not of the DOF. Re-executing both the translational and yaw identification pipelines with identical seed, data, and toolbox version reproduces all nine parameters to 0.0 absolute difference (bit-exact), and the SE(2) gate verdict for all three DOFs is unchanged under a $\pm 21\%$ perturbation of T_d to its neighboring integer-step quantization (a robustness check, since the simulator's transport delay is in fact continuous and unquantized).

E. A joint behavioral calibration does not beat per-DOF on translation

As a negative control, a joint 9-parameter behavioral calibration (Bayesian optimization with a multi-horizon SE(2) cost, validated against the high-fidelity simulator by a ranking proxy, Spearman $\rho = 0.999$) was evaluated on the same hold-out. It beats the ideal baseline (pooled $-0.098\,\text{m}$, $p = 7.6 \times 10^{-10}$) but does *not* beat the per-DOF translational model (pooled $+0.0041\,\text{m}$, $p = 0.040$, wrong direction): it

matches rotation and loses translation, and is not promoted. This supports the per-DOF grey-box choice: closed-loop logs do not sustain a joint behavioral fit that improves translational fidelity over per-DOF identification.

V. Discussion

A. Identifiability is a property of the data, not the DOF

The central finding of this paper is not that yaw is harder to model than translation, but that the *same* model structure, fit with the *same* tooling, gives opposite verdicts depending on the excitation regime of the data. Fit to closed-loop competition logs, the yaw P1D model fails on every axis: physical bounds are marginal, hold-out fit is below threshold, and the $\mathrm{SE}(2)$ gate says it is worse than a no-dynamics placeholder. Fit to a small dedicated open-loop collection with the same structure, it passes every pre-registered gate, with a gain statistically indistinguishable from unity. Framed as a property of ω, this would read as "translation is identifiable, yaw is not." Framed as a property of the *data*, it reads instead as: closed-loop competition commands do not excite yaw enough to identify it, and 94% of the commanded yaw-rate samples in our logs sit below the robot's own actuation threshold, even on the motion label richest in rotation. The translational channels never faced this problem because the same logs excite v_x, v_y persistently (Sec. III-D); nothing about the identification pipeline distinguishes the DOFs, only the data does.

B. Correlation bias as the mechanism, not a coincidence

The closed-loop gain estimate ($K_\omega \approx 1.3$–1.35) is not merely imprecise; it is systematically wrong in the same direction across two independent closed-loop attempts (a direct high-rotation fit and a regime-restricted linear fit), and the open-loop estimate ($K_\omega = 1.0016$, CI excludes 1.3) refutes it as a plant property. The reading we favor is that, under closed-loop play, a human or tactical controller continuously corrects the trajectory, so the commanded yaw rate correlates with unmodeled disturbances – surface friction, contact, tactical override – in a way that inflates the apparent gain of a model fit to command-to-velocity data alone. This is a textbook closed-loop identification hazard closed-loop system identification / lack of persistency of excitation under feedback control, and it is consistent with the direction of the earlier failure: the closed-loop candidate's gain > 1 specifically *worsened* free-run position by amplifying accumulated heading drift (Sec. IV-D), which is what an inflated gain would do.

C. Where the twin beats rSim, and why

On translation-dominated windows the twin's advantage over the ODE-based rSim engine grows with horizon rather than shrinking (Sec. IV-C), which is the signature of a dynamical advantage rather than a favorable initial condition: the transport delay and first-order lag in Eq. 1 capture a short-horizon response that a rigid-body ODE engine without an explicit delay term structurally cannot. The gap widens further under Mode B (no warm-up), where rSim's rest-start penalty compounds; Mode A is deliberately conservative and is used as the primary result for this reason.

D. Why the joint 9-parameter calibration does not help translation

The negative control (Sec. IV-E) shows that jointly calibrating all nine parameters against a multi-horizon $\mathrm{SE}(2)$ cost does not improve translational fidelity over the per-DOF fit, even though the joint cost function has direct access to the same closed-loop data and a cheap-to-evaluate proxy. This is evidence *for* the per-DOF decomposition used throughout this paper: a joint behavioral objective, fit to the same excitation-poor closed-loop logs, cannot substitute for identifying each channel against the regime where it is actually excited – the same lesson that motivated the dedicated open-loop yaw collection.

E. A methodological lesson beyond this robot

We frame the yaw result as a limit of identification from competition logs, not as a repair of an earlier mistake. RoboCup teams generally have abundant closed-loop competition logs and comparatively few dedicated, open-loop identification experiments RoboCup SSL logging practice / general scarcity of dedicated system-identification data collection in robot soccer. The pattern here – a per-DOF model that looks unidentifiable under one data regime and identifies cleanly under another, with the discrepancy explained by a quantifiable actuation threshold rather than by the model structure – is not specific to this robot's yaw channel. It suggests that "this DOF is not identifiable" claims drawn from competition logs alone should be treated as claims about the available excitation, not about the system, until checked against a regime that persistently excites the DOF in question. We keep this claim calibrated to e_{pos} throughout; heading error is reported but does not carry a verdict where the population does not support one (Sec. IV-B), and we do not generalize the translational result to rotation-dominated regimes (Sec. VI).

VI. Limitations and Threats to Validity

a) Small sample for the yaw identification: The open-loop yaw fit uses only six events in the linear regime (four training, two hold-out) from a single collection session. Amplitudes were not swept finely between 0.5 and $1.0\,\mathrm{rad/s}$ – the robot was not available for a follow-up session within the study timeline – so the commanded-yaw dead-zone is reported as an interval, 0.58–$0.60\,\mathrm{rad/s}$, never as a point estimate; the empirically observed boundary is the bracket $(0.5, 1.0]\,\mathrm{rad/s}$, not a value inside it. The small sample is mitigated, not eliminated, by pre-registering the acceptance criteria before estimation (Sec. III-E) and by confirming bit-exact determinism of the fit under re-execution (Sec. IV-D); it is not mitigated by cross-session or cross-robot replication, which we do not have.

b) Single robot, single session: Both the open-loop yaw collection and, more broadly, the dataset used for v_x, v_y identification and evaluation come from a single robot and a single data-collection context. We do not report cross-robot or cross-session variability for any of the three DOFs.

c) Rotational fidelity of the twin remains unverified: Identifying ω resolves the *parameter*, but the twin-vs-baseline $\mathrm{SE}(2)$ gate on competition-like data still returns no verdict for mixed and rotation-dominated windows, because the hold-out has essentially no such windows (0 mixed, 1 rotation-dominated out of 265), not because of a placeholder gate (Sec. IV-B). This is a distinct and more fundamental limitation than the one it replaces: the rotational fidelity of the digital twin, as opposed to the yaw parameter estimate itself, is not validated against a distribution resembling competition play.

d) An unexplained kinematic constant: The wheel-to-contact-radius ratio implied by the firmware's live telemetry ($R/r \approx 2.61$) differs from the ratio measured from the robot's CAD model ($R/r \approx 2.699$) by about 3.3%. The likely explanation is a version drift between the firmware source inspected and the binary actually flashed during data collection, but this is not confirmed, and we report the discrepancy rather than silently pick one source.

e) An untraceable legacy constant: An earlier body-frame yaw dead-zone threshold of $0.3\,\mathrm{rad/s}$, used in prior versions of this pipeline to estimate how much of the competition logs' yaw commands were masked by actuation limits, has no recoverable provenance: it is present, unexplained, at the root commit of this project's history. We report it as inherited and superseded by the per-wheel electrical threshold characterized in this paper, not as a value we can justify or trace further.

f) Whiteness test is informative, not a gate: The Ljung–Box test rejects residual whiteness for the yaw fit, in both the closed-loop and open-loop attempts. As pre-registered, this is not treated as an acceptance gate: the measured signal is a Savitzky–Golay-smoothed derivative of vision-derived heading, which colors the residual noise by construction, and a large sample size makes rejection likely even when the identified gain is otherwise correct. We report it for transparency, not as evidence against the fit.

g) Baseline fairness: rSim is evaluated at its default parameters, not re-tuned to this robot; a re-tuned ODE simulator could narrow the observed gap. The two initial-condition fairness modes (A/B, Sec. III-G) bracket rSim's rest-start disadvantage but do not eliminate the asymmetry that the twin has an explicit transport-delay term and rSim does not.

h) Scope of the evaluation data: All results come from a single dataset and a single hold-out split; we do not have a second dataset, court, or firmware version to test cross-log generalization. Hold-out events are drawn from the same operating regime as the training events, so generalization to unseen maneuvers or speeds is not tested. The native time grid is irregular, and the initial condition for free-run replay relies on vision-derived velocity, which is noisy; both are mitigated (resampling, filtering) rather than eliminated.

i) Reproducibility of the rSim baseline: The rSim comparison was produced in a Linux environment with a pre-built ODE dependency that does not compile in our default Windows environment; reproducing Sec. IV-C requires documenting and matching that environment.

VII. Conclusion and Future Work